

Week 4 Weds 9/23

→ 10:15 missed bits of lecture

$$\vec{F}_{\text{ext}} = M \ddot{\vec{R}}$$

Conservation of Momentum

$$\frac{d}{dt} \vec{P}_{\text{total}} = 0 \quad \text{if no external forces}$$

Example 4.8 Spring Gun



$$t=0: \quad P_x = 0$$

$$t>0: \quad P_x = -Mv_f + m(v_0 \cos \theta - v_f) = 0$$

$$\hookrightarrow v_f = \frac{mv_0 \cos \theta}{m+M}$$

this is this contribution from relative motion

momentum conserved
↓
key insight because the particle moves relative to Block, which is already moving

Impulse (ch 4.6)

→ We can integrate $\vec{F} = \frac{d\vec{P}}{dt}$

$$\Delta \vec{P} = \int_0^{\delta} \vec{F} dt = \vec{P}(t) - \vec{P}_0$$

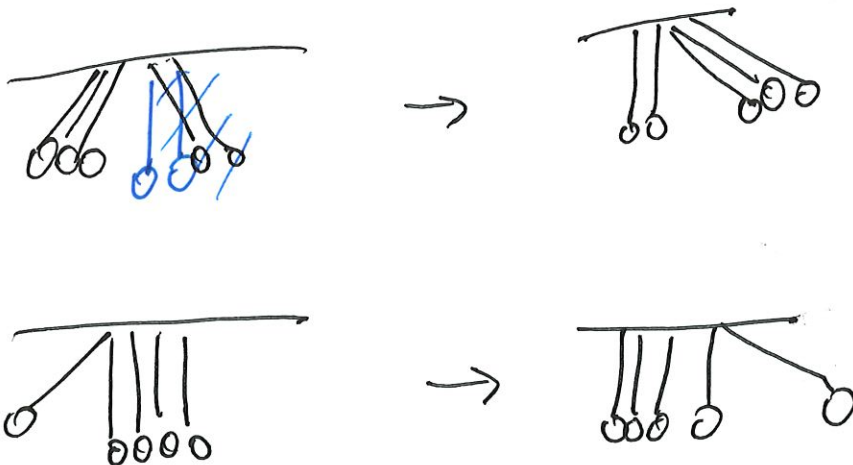
this is the Impulse

ex (4.10)

Bouncing Ball vs Egg

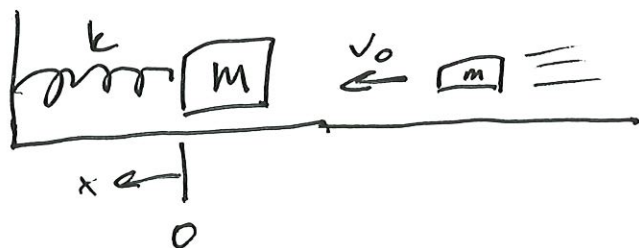


Newton's Cradle!



Ex 4.9

Measuring Speed of Bullet



"Suppose you didn't get the grant and you can't buy that first camera."

→ How do we measure the bullet on a budget?"

Conserve Momentum!

→ $t=0$: $P_x = mv_0$

Momentum Conserved
 b/c $\sum p^{\text{ext}} = 0$
 there are no forces

~~$t=0$~~ : $P_x = v_i (m+M)$

$t=0+\epsilon$
 ↑
 small #

↳ this "initial" is after impact
 but before force of spring compression

$t > \epsilon$: $(m+M)\ddot{x} = -kx$ } Momentum no longer conserved (!)
 $-kx = F_{\text{spring}}$ is ext. force

→ What's the solution? 2nd order ODE
 → sine and cosine!

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

→ $\omega = \sqrt{\frac{k}{m+M}}$

, $A=0$, $B = v_i / \omega$

← dimensions?

$x(t) = \frac{v_i}{\omega} \sin(\omega t) \Rightarrow \max |x| = \frac{v_i}{\omega}$

We know from Momentum conservation

$$v_i (m+M) = m v_o$$
$$\downarrow$$
$$v_i = v_o \left(\frac{m}{m+M} \right)$$

then

$$\max |x| = \frac{v_i}{\omega} = v_o \left(\frac{m}{m+M} \right) \sqrt{\frac{m+M}{k}}$$
$$= \frac{m v_o}{\sqrt{k(m+M)}}$$

□

If we measure the displacement
(and if we know m, M, k)

We can calculate the Bullet's speed v_o .